

4. Discussion and Future Work

Choice of Steering Vectors. An analysis of individual steering directions reveals substantial variability in their effects. Some vectors induce more severe degradations in generation quality, while others are more easily detectable. Steering directions also differ in their impact on the representation space at detection time, as measured by cosine similarity in Figure 5. We hypothesize that these differences arise from varying degrees of orthogonality to the model’s natural activation trajectory in a given context. This heterogeneity suggests that selecting or optimizing subsets of steering vectors that balance detectability and generation quality could further improve performance.

Multi-bit Watermarking. As proposed in Section 3.2, superimposing multiple sparse vectors with disjoint supports theoretically enables a high-capacity, multi-bit watermark. In such a scheme, the presence or absence of a specific steering vector encodes a single bit of the unique identifier. Achieving this requires further research to verify that the linear superposition of steering signals remains recoverable within the model’s activation space without destructive interference.

Interpretability of Random Steering. To the best of our knowledge, utilizing sparse random vectors is a novel approach in activation engineering, specifically for watermarking. While our metrics indicate negligible degradation in quality, the semantic nuances induced by these perturbations remain unclear. We could not identify obvious artifacts in the text output, yet a subtle shift in distribution is inevitable to allow detection. Future work must rigorously analyze these impacts, potentially leveraging automated annotation by strong judge models or interpreting the steered features via sparse autoencoders for monosemantic feature discovery (Templeton et al.). Such interpretability is a prerequisite for deployment, ensuring that the watermark does not inadvertently introduce safety risks or biases while helping to design more semantically neutral steering vectors.

Cross-evaluation results displayed in Appendix G reveal a hard architectural boundary: detection remains strong within the same family but degrades to chance level when text is generated by another architecture, regardless of dataset. This mirrors observations of the *subliminal learning* phenomenon reported by Cloud et al. (2025), where implicit behavioral traits in teacher-generated data transfer to a student only when both models share the same base architecture. We interpret both findings as evidence of architecture-specific activation structure: steering vectors leave activation traces that are invisible at the semantic level but geometrically coherent within a model family, and do not generalize beyond it.

Robustness and Secrecy. While we demonstrated robustness against strong paraphrasing, the watermark’s resilience to broader attacks, such as recursive translation, adversarial perturbations, text degradation, or format changes, remains to be characterized. Furthermore, the security of the method relies on the secrecy of the steering configuration. Specifically, if an attacker can recover the secret steering vector and target layer, the watermark could be trivially reproduced to spoof the identity of the steered LLM. To mitigate this risk, future work could explore dynamic steering strategies in which the vector rotates or evolves during generation according to a pseudo-random schedule. Such a moving target would significantly complicate the recovery of the watermark signal without the corresponding seed.

5. Limitations

Detection Overhead. Unlike inference-time watermarks, signal injection incurs no computational cost in our method. However, detection requires white-box access to the model and a forward pass to extract activations from the text, which incurs a non-negligible computational cost. In our setup using an RTX A5500, collecting the model’s activations takes time equivalent to 21% of the time needed to autoregressively generate the same text.

Quality Evaluation Limitations. To obtain a more accurate evaluation of textual quality than perplexity, we relied on an externally fine-tuned quality assessment model. However, this evaluator exhibits biases toward structural formatting, such as penalizing markdown syntax regardless of content quality. Theoretically, degraded texts may still be assigned high scores, potentially obscuring small performance losses in steered outputs. Despite these limitations, we maintain that measuring the *relative* quality difference between steered and unsteered outputs provides a valid proxy for steering-induced degradation, as formatting and other confounders remain constant across conditions.

6. Related Work

Recent work on mechanistic interpretability shows that transformer activations encode linearly recoverable signals for high-level concepts. In the context of truthfulness, simple probes of hidden states can predict the veracity of a statement even when the model’s final answer is incorrect, suggesting concept directions that are partially disentangled from generation (Marks & Tegmark, 2023; Azaria & Mitchell, 2023).

Several approaches use internal representations for detecting AI-generated text. RepreGuard (Chen et al., 2025) separates human and model text using observer-model hidden states, while Yu et al. (2024) train supervised classifiers on acti-

variations from an auxiliary encoder. These works primarily target robust, prompt-agnostic detection of arbitrary LLM-generated text, rather than self-recognition. SAEMark (Yu et al., 2025) uses SAE features to guide a rejection-sampling process for watermarking, achieving robust detection at the cost of significant computational overhead during generation.

Closest to our setting, Ackerman & Panickssery (2025) study whether instruction-tuned models can recognize their own outputs and extract residual-stream directions via contrastive pairs, but report performance that does not consistently beat perplexity-based heuristics. Separately, Kuznetsov et al. (2025) show that SAE-based features can improve general-purpose AI-GTD. They use feature steering to analyze the nature of the features responsible for detection and causally link the intervention to detectability.

Conclusion

In this work, we investigated the self-recognition capabilities of LLMs and introduced a random steering mechanism to enable the attribution of generated text. Our results show that LLMs exhibit strong self-recognition capabilities, encoded as linearly separable signals in their internal activations. These signals are distributed across the network, robust to semantic shifts, and enable attribution without relying on perplexity.

Furthermore, a simple inference-time intervention enables robust source attribution while incurring minimal degradation in text quality. Importantly, we find that sparse interventions in the activation space lead to lower quality loss than dense noise injection for comparable detectability. Moreover, by operating directly on model activations, our approach avoids reliance on computationally intensive sparse autoencoder pipelines while still capturing high-level properties of the inference process.

Future work may extend this analysis to the design of a seamless, resource-efficient watermarking system, as well as to systematic comparisons with existing approaches. More broadly, we hope this work encourages further investigation into the inference-time dynamics of LLMs and into how internal representations can be leveraged to ensure reliable and transparent model behavior.

Impact Statement

This paper examines whether LLMs encode a reliable self-recognition signal in their internal activations and leverages it to enable watermarking and attribution at the level of a model or a steered variant. Such capabilities could support provenance, auditing, and accountability for AI-generated text, but they also amplify known risks. In particular, water-

marking for attribution can raise privacy concerns if signatures directly link to specific deployments or users.

We identify no substantive risks associated with the publication of our method, and our contribution is purely methodological. The threat models we evaluate are standard in the watermarking literature and can be executed using publicly available tools. Consequently, disclosing our method does not introduce any new adversarial capabilities beyond those already well known in existing watermarking frameworks.

Author Contributions

TA and JS both contributed significantly to this work. TA led the steering experiments and writing, JS led the self-recognition experiments, under the general direction of GW.

Acknowledgements

TA, JS and GW were supported by the Federal Ministry of Education and Research of Germany (BMBF) in the programme of “Souverän. Digital. Vernetzt.”, joint project “AIgenCY : Chances und Risks of Generative AI in Cybersecurity”, project identification number 16KIS2013. GW was also supported by BMBF joint project “6G- RIC: 6G Research and Innovation Cluster”, project identification number 16KISK025.

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